VY TRUONG

SOFTWARE DEVELOPMENT STUDENT

Brisbane, QLD | 0439 119 687
vivian.trg3@gmail.com | github.com/vy-trg | linkedin.com/in/vy-trg | vy-trg.github.io/portfolio
Student visa: 48 hours per fortnight during semester, full-time during breaks | Available full-time until Feb 2027

PROFILE

Penultimate-year Bachelor of Information Technology (Software Development) student at Griffith University. I write tested, maintainable code in Java, JavaScript and Python, from game logic and a game-playing AI to responsive websites, relational databases and Power BI dashboards. Busy hospitality roles have taught me to stay accurate under pressure and communicate clearly. I'm looking for a software development internship, WIL placement or graduate role.

EDUCATION

Bachelor of Information Technology (Software Development) **Mar 2025 – Nov 2027 (expected)**
Griffith University, Brisbane

TECHNICAL SKILLS

- Languages & frameworks: Java, JavaScript, Python, SQL, HTML, CSS; JavaFX, JUnit 5, Mockito, Python unittest, p5.js, NumPy, Matplotlib
- Testing & design: unit testing (positive and negative cases, parameterized tests), stubs and mocks, branch coverage, CI with GitHub Actions; object-oriented programming, design patterns (State, Factory, Singleton, Facade), MVC, layered architecture, REST APIs, relational database design (ER diagrams), data modelling (star schema)
- Process & tools: SDLC, Agile, requirements and use cases, WBS, Gantt charts, risk registers; Git, GitHub, Maven, IntelliJ IDEA, VS Code, Chrome DevTools, MySQL, Microsoft Power BI (Power Query, DAX), Excel, draw.io

SELECTED PROJECTS

JavaFX Tetris | Java 21, JavaFX, JUnit 5, Mockito, Maven, GitHub Actions **2026**
github.com/vy-trg/javafx-tetris
3-person university team project, graded 96/100

- Built the core gameplay engine: the board, seven tetromino types (abstract class, interface and factory), rendering, keyboard controls and row clearing, delivered through a feature branch and a reviewed pull request.
- Restructured the game into Model–View–Controller with the State pattern, so all game rules run without the UI and can be unit tested.
- Developed a heuristic AI player that tests every rotation and column on a copy of the board, scores each result on height, holes, bumpiness and lines cleared, and plays through the same move methods as a human.
- Built two-player mode where each side can be a human, the AI or a networked player, using a thread-safe shared piece sequence; fixed a JavaFX threading bug that stopped the Game Over dialog appearing.
- Wrote all 21 unit tests, including parameterized tests across three board sizes, a stub, Mockito mocks and an 8-thread check of the shared piece sequence, and set up GitHub Actions to run them on every push.

EnergyCheck | HTML, CSS, JavaScript **2026**
github.com/vy-trg/energycheck | Live demo
5-person university team project, High Distinction

- As team leader and Design Thinking Lead, guided the team through empathy research, prototyping and user testing to build a 10-page responsive website for Australian renters.
- Built the shared stylesheet with CSS custom properties, a responsive grid and dark mode, so every team member's pages looked and behaved the same.
- Wrote the site's JavaScript (saved dark-mode preference, high-contrast mode, mobile menu, chart tooltips) and fixed accessibility issues with ARIA attributes.

Queensland Railway Interactive Map | JavaScript, p5.js, GeoJSON **2025**

github.com/vy-trg/qld-railway-map | Live demo

University coursework (Creative Coding), graded 40/40

- Built an interactive map from real Queensland railway station data in GeoJSON, with hover tooltips showing each station's name, coordinates and operational status.
- Drew each train line in its own colour, scaling geographic coordinates accurately onto a real background map.

Kakadu Kitchen Database System | SQL, MySQL, ERD **2025**

github.com/vy-trg/kakadu-kitchen-database

University team project (Data Management)

- Designed and built a normalised 9-table MySQL database from an ER diagram, covering products, partners, partnerships and packaging, with primary and foreign keys and sample data.
- Wrote eight analytical SQL queries using joins, aggregations and subqueries to answer business reporting questions.

Power BI Dashboards and Reports | Power BI, Power Query, DAX, Excel **2026**

University coursework (Business Intelligence)

- Cleaned and reshaped raw Excel data in Power Query (headers, data types, unpivoting, removing aggregate rows) and chose the right aggregation for each metric, such as averaging rather than summing GDP per capita.
- Built a star-schema data model with a dedicated date table, and found and fixed a relationship error caused by duplicate product IDs.
- Wrote DAX measures with CALCULATE, FILTER and ALL, including financial-year-to-date sales for Australia's July–June financial year.
- Designed dashboards with matrices, KPI cards, slicers, drill-through and conditional formatting, and wrote a 1,500-word report turning the results into specific business recommendations.

OTHER PROJECTS

- REST Countries app: JavaScript, REST API, search and filter, light/dark theme — Code | Live demo
- Space Shooter game: p5.js, four scenes, collision detection, JSON leaderboard — Code | Live demo
- Python data analysis: NumPy, Matplotlib, savings model, rainfall data — Code

WORK EXPERIENCE

Service Staff, Hikari Ramen, Orion Shopping Centre, Springfield QLD **Jun 2025 – Present**

- Take dine-in, takeaway and phone orders and process POS payments during busy lunch and dinner peaks.
- Handle phone order changes, missing items and group bookings, reading details back to prevent mistakes.
- Resolve customer complaints calmly, and escalate to the manager when an issue is beyond my role.
- Work closely with kitchen staff to keep orders accurate and moving during high-volume shifts.

Service Staff, Dadam Korean BBQ, Orion Shopping Centre, Springfield QLD **Mar – Apr 2026**

- Coordinated with kitchen staff and monitored table equipment to keep service running smoothly.
- Served customers in a fast-paced restaurant with attention to speed, accuracy and teamwork.

Mathematics Tutor (Freelance), Da Nang, Vietnam **May – Oct 2024**

- Tutored students one-on-one at different levels, adapting lesson plans to each student's needs.
- Tracked progress and explained concepts step by step to improve students' understanding and results.

LANGUAGES

English (fluent), Vietnamese (native)